

LAWRENCIUM MODEL HANDOFF

Checking and Evaluating the 1M-row QLoRA Model

Verified completion evidence, exact loading steps, a reproducible base-model comparison, and publication-safe next steps

Training result: COMPLETED - 12,090/12,090 steps - exit 0

Hardware: Four NVIDIA A40 GPUs

Dataset: 1,000,000 rows - one epoch

Updated: August 31, 2026

What finished

The production pilot completed one epoch over 1,000,000 training rows using four NVIDIA A40 GPUs. All 12,090 optimizer steps finished. The final reported training loss was 0.06389, and held-out evaluation loss fell monotonically from 0.10089 at step 500 to 0.04937 at step 12,000. The run saved a 168 MB LoRA adapter and retained checkpoints 11,500, 12,000, and 12,090.

This proves the training pipeline ran correctly and optimized held-out loss. It does not by itself prove that molecular generations or synthesis plans are chemically better. That requires output-level comparison against the untouched base model and chemistry-specific evaluation.

Functional generation check completed

Lawrencium job [25428169](#) loaded the untouched base model and the completed QLoRA adapter sequentially on one A40, generated all 25 requested outputs, and completed with exit code 0. The fixed sample contained nine SmileyLlama, nine SynLlama, and seven LinkLlama records. Both models returned nonempty output for 25/25 prompts, and none of the base outputs was identical to its QLoRA output.

As a basic format-similarity diagnostic, normalized character-sequence similarity to the saved reference averaged 0.0508 for the base outputs and 0.5500 for the QLoRA outputs. Exact normalized reference matching was 0/25 for the base model and 1/25 for QLoRA. Mean output length decreased from 685.6 to 240.5 characters. These numbers show that the adapter materially changed model behavior toward the training response format.

This is intentionally labeled a **functionality check**, not a quality score: the 25 records were sampled from the staged training file, character similarity is not a chemistry metric, and the three source tasks require different

evaluators. The raw comparison is preserved in `evaluation-2026-08-31/base_vs_qlora.jsonl`, with a readable companion at `evaluation-2026-08-31/base_vs_qlora_preview.md`.

Fast status check

Log in to Lawrencium, then run:

```
sacct -j 25306635 --format=JobID,JobName,State,ExitCode,Elapsed,Start,End -X
tail -40 ~/lawrencium/qlora_1m_25306635.out
du -sh /global/scratch/users/$USER/runs/qlora_1M_20260827
ls -lh /global/scratch/users/$USER/runs/qlora_1M_20260827
```

Expected evidence:

```
State: COMPLETED
ExitCode: 0:0
Steps: 12090/12090
Training Completed!!!
adapter_model.bin: approximately 168 MB
```

Run a reproducible base-versus-adapter comparison

The branch includes `40_compare_adapter.py` and `40_compare_adapter.sbatch`. The job uses one A40 and loads the base and fine-tuned models sequentially. It generates deterministic outputs for the same 25 records and writes both raw JSONL and a readable Markdown preview.

From the local repository, transfer the scripts without relying on the login node's full `/tmp`:

```
tar czf - docs/lawrencium/40_compare_adapter.py \
docs/lawrencium/40_compare_adapter.sbatch | \
ssh lrc 'tar xzf - --strip-components=2 -C ~/lawrencium'
```

On Lawrencium:

```
cd ~/lawrencium
sbatch 40_compare_adapter.sbatch
queue -u $USER
```

After completion:

```
cat /global/scratch/users/$USER/runs/qlora_1M_20260827/evaluation/base_vs_qlora_preview.md
wc -l /global/scratch/users/$USER/runs/qlora_1M_20260827/evaluation/base_vs_qlora.jsonl
```

The comparison uses greedy decoding (`do_sample=False`) and a fixed seed so the base and adapter receive identical prompts. The sampled file is the staged training subset, so this is a functional smoke comparison, not an unbiased test-set estimate. For a paper, replace `--data` with a truly held-out JSONL that was excluded before training.

Load the adapter in Python

The saved artifact is a LoRA adapter, not a standalone 8B model. Load the same base checkpoint first and then attach the adapter:

```

import os
import torch
from peft import PeftModel
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig

base_id = "NousResearch/Meta-Llama-3.1-8B-Instruct"
adapter_path = os.path.expandvars(
    "/global/scratch/users/$USER/runs/qlora_1M_20260827"
)

quant = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.bfloat16,
    bnb_4bit_use_double_quant=True,
)
tokenizer = AutoTokenizer.from_pretrained(base_id)
base = AutoModelForCausalLM.from_pretrained(
    base_id, quantization_config=quant, device_map="auto"
)
model = PeftModel.from_pretrained(base, adapter_path)
model.eval()

```

Keep model loading on a GPU compute node.

What to measure next

Use a held-out set with source labels and compare the untouched base model and QLoRA adapter using identical prompts and decoding:

1. Molecule generation: valid-SMILES rate, property error, uniqueness, and scaffold diversity.
2. Retrosynthesis: parse success, exact route validity, permutation-aware route validity, and analog-aware route validity.
3. Instruction following: schema compliance and required-field completion.
4. Human review: blinded assessment of a stratified sample by a synthetic chemist.
5. Efficiency: generation latency, GPU memory, and failure rate.

Do not choose the best-looking checkpoint using the test set. Pick a checkpoint using validation loss or a predeclared development metric, then report the test set once. Checkpoint 12,000 has the lowest recorded evaluation loss among the regular 500-step checkpoints; checkpoint 12,090 is the final training state.

Recovery and preservation

Scratch is not backed up. Preserve the adapter, configuration, tokenizer, and trainer state before relying on them for publication:

```
RUN=/global/scratch/users/$USER/runs/qlora_1M_20260827
tar -C "$(dirname "$RUN")" -czf "$HOME/qlora_1M_20260827_adapter.tar.gz" \
  qlora_1M_20260827/adapter_model.bin \
  qlora_1M_20260827/adapter_config.json \
  qlora_1M_20260827/config.json \
  qlora_1M_20260827/tokenizer.json \
  qlora_1M_20260827/tokenizer_config.json
sha256sum "$RUN/adapter_model.bin" "$RUN/adapter_config.json"
```

Recorded SHA-256 checksums:

```
adapter_model.bin    9e3896eb80eba3397911246aaea1eecbc2480059abcd747b2030de8beaf294d4
adapter_config.json  f6c4dc78a255874d6b5092ba30d41b6f171fb028ec57354cc5db15a60ebab981
```